Note: Every Hermitian operator on fin. dim inn. prod space is diagonalizable by it's eigenvectors.
In particular, in $\mathbb{R}$, it is necessary to satisfy the condition, Normal

Spectral Theorem

Let $T: V \to V$ be a Hermitian or Normal (in $\mathbb{R}$).
Let $\lambda_1, \cdots, \lambda_k$ be distinct eigenvalues for T, and $W_i = \ker(T - \lambda_i I)$, $1 \le i \le k$.
Let T_i be an orthogonal projection onto W_i. Then,

1). $V = W_1 \oplus \cdots \oplus W_2$

2). $W_i^{\perp} = \bigoplus_{j \neq i} W_i$

3). $T_i \circ T_j = \delta_{ij} T_i$

4). $I = T_1 + \cdots + T_k$

5) $T = \lambda_1 T_1 + \cdots + \lambda_k T_k$.

Corollary. In $\mathbb{C}$,

T is normal if and only if $T^* = g(T)$ for some polynomial g.

proof. Let $g(x) = \sum_{i=1}^{k} \left[\left(\prod_{j \neq i} (x - \lambda_j) \right) \Big/ \left(\prod_{j \neq i} (\lambda_i - \lambda_j) \right) \right] \cdot \overline{\lambda_i}$.

Then, $g(\lambda_i) = \overline{\lambda_i}$, so $g(T) = \sum \overline{\lambda_i} T_i = T^*$.

Corollary. In $\mathbb{C}$,

T is unitary if and only if T is normal and every eigenvalue λ of T has $|\lambda| = 1$.

Corollary. In $\mathbb{C}$

T is unitary if and only if T is normal and every eigenvalues are real.

⌈Thm. Singular Value decomposition.

Let V and W be fin. dim inner product spaces, and $T: V \to W$ be a rank r linear operator.
Then, there exist unique positive scalar $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0$ and orthonormal basis $\{v_1, v_2, \cdots, v_n\}$ and $\{u_1, \cdots, u_m\}$ for V and W.

$$T(v_i) = \begin{cases} \sigma_i \cdot u_i & (1 \leq i \leq r) \\ 0 & (i > r) \end{cases}$$

Proof. T^*T is semi-positive, because $\langle T^*T\alpha, \alpha \rangle = \|T\alpha\|^2$. So all eigenvalues are positive.
And, rank T^*T = rank T, because given $\alpha \in V$, $T(\alpha) = 0 \Rightarrow T^*T(\alpha) = 0$, so $\ker T \subseteq \ker T^*T$.
Conversely, if $T^*T(\alpha) = 0$, then $0 = \langle 0, \alpha \rangle = \langle T^*T(\alpha), \alpha \rangle = \|T(\alpha)\|^2 = 0$, so $\ker T^*T \subseteq \ker T$.
By the dimension Theorem, the rank are same.⌋

Now, since T^*T is Hermitian, there is an orthonormal eigenvectors basis for V, $\{v_1, \cdots, v_n\}$, each v_i's corresponding to positive eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$.
For each $1 \leq i \leq r$, define

$$\sigma_i = \sqrt{\lambda_i} \quad \text{and} \quad u_i = \frac{1}{\sigma_i} \cdot T(v_i).$$

Then, for each $1 \leq i, j \leq r$,

$$\begin{aligned} \langle u_i, u_j \rangle &= \left\langle \frac{1}{\sigma_i} \cdot T(v_i), \frac{1}{\sigma_j} \cdot T(v_j) \right\rangle \\ &= \frac{1}{\sigma_i \sigma_j} \cdot \langle \underbrace{T^*T(v_i)}_{= \lambda_i v_i}, v_j \rangle \\ &= \frac{\sigma_i^2}{\sigma_i \sigma_j} \langle v_i, v_j \rangle = \delta_{ij}. \end{aligned}$$

Now, extend $\{u_1, \cdots, u_r\}$ to $\{u_1, \cdots, u_r, \cdots, u_m\}$ using Gram-Schmidt, proved.⌋

⌈Corollary. Any $A \in M_{m \times n}(\mathbb{C})$, there exists unitary $U \in M_{m \times m}$, $V \in M_{n \times n}$ s.t.

$$A = U \Sigma V^* \quad \text{where } \Sigma = \begin{pmatrix} \sigma_1 & & & \\ & \ddots & & 0 \\ & & \sigma_r & \\ 0 & & & \ddots \; 0 \end{pmatrix}, \quad r = \text{rank} A.$$

Proof. Let $\beta = \{v_1, \cdots, v_n\}$ be an orthonormal basis for $\mathbb{C}^n$ consisting by eigenvalues of A^*A, and for this eigenvalues λ_i, put $u_i = (\sqrt{\lambda_i})^{-1} \cdot Av_i$, and $\beta' = \{u_1, \cdots, u_r, \cdots, u_m\}$, the $u_{r+1}, \cdots, u_m$ are obtained by Gram-Schmidts process. Then, for standard basis α, α',

$$[L_A]_\beta^{\beta'} = [I_m]_{\alpha'}^{\beta'} [L_A]_\alpha^{\alpha'} [I_n]_\beta^\alpha = \Sigma, \text{ so Proved.}$$

Example.

Let $V = P_2(\mathbb{R})$ and $W = P_1(\mathbb{R})$, with inner product

$$\langle f, g \rangle = \int_{-1}^{1} f(t) \cdot g(t)\, dt.$$

Consider the differentiation $D : V \to W : f \mapsto Df$.

Using Gram-Smidt Process, find an orthonormal basis for V as;

$$\alpha_1 = 1$$

$$\alpha_2 = x - \frac{\langle x, 1 \rangle}{\|1\|^2} \cdot 1 = x$$

$$\alpha_3 = x^2 - \frac{\langle x^2, 1 \rangle}{\|1\|^2} \cdot 1 - \frac{\langle x^2, x \rangle}{\|x\|^2} x = x^2 - \frac{1}{3} - 36 \cdot x \cdot 0 = x^2 - \frac{1}{3}.$$

Now, by normalizing, we obtain

$\beta = \left\{ \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} x, \sqrt{\frac{5}{8}} (3x^2 - 1) \right\}$, $\beta' = \left\{ \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} x \right\}$ are basis for V and W.

So, $[D]_{\beta}^{\beta'} = \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix}$

Now, $[D^* D]_{\beta} = \begin{pmatrix} 0 & 0 \\ \sqrt{3} & 0 \\ 0 & \sqrt{15} \end{pmatrix} \begin{pmatrix} 0 & \sqrt{3} & 0 \\ 0 & 0 & \sqrt{15} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 15 \end{pmatrix}$

So, the singular values are $\sqrt{15} \geq \sqrt{3} \geq 0 \cdots$

Let $A=\begin{pmatrix}1&1&-1\\1&1&-1\end{pmatrix}\in M_{2\times 3}(\mathbb{R})$ be given.

Then, $A^*A=A^TA=\begin{pmatrix}1&1\\1&1\\-1&-1\end{pmatrix}\begin{pmatrix}1&1&-1\\1&1&-1\end{pmatrix}=\begin{pmatrix}2&2&-2\\2&2&-2\\-2&-2&2\end{pmatrix}$.

$$\begin{vmatrix}2-t&2&-2\\2&2-t&-2\\-2&-2&2-t\end{vmatrix}=(2-t)\cdot((2-t)^2-4)-2\cdot(4-2t-4)$$
$$-2\cdot(-4+4-2t)$$
$$=(2-t)\cdot(-4t+t^2)-2\cdot(-2t)-(-2t)$$
$$=-\cancel{8t}+2t^2+4t^2-t^3+\cancel{8t}$$
$$=-t^3+6t^2=-t^2(t-6).$$

$$\begin{pmatrix}2&2&-2\\-2&-2&2\\0&0&0\end{pmatrix}\to\begin{pmatrix}1&1&-1\\-1&-1&1\\0&0&0\end{pmatrix}\to\begin{pmatrix}1&1&-1\\0&0&0\\0&0&0\end{pmatrix}$$

$$\begin{pmatrix}1&1&-1\\0&0&0\\0&0&0\end{pmatrix}\begin{pmatrix}x\\y\\z\end{pmatrix}=0 \qquad \begin{pmatrix}-4&2&-2\\2&-4&-2\\-2&-2&-4\end{pmatrix}$$

$\ker A^TA=\operatorname{span}\left\{\begin{pmatrix}1\\1\\2\end{pmatrix},\begin{pmatrix}1\\-1\\0\end{pmatrix}\right\}$,

$\ker(A^TA-6I)=\operatorname{span}\left\{\begin{pmatrix}1\\1\\-1\end{pmatrix}\right\}$.

Now, let $V=\left(\frac{1}{\sqrt3}\cdot\begin{pmatrix}1\\1\\-1\end{pmatrix}\ \frac{1}{\sqrt2}\begin{pmatrix}1\\-1\\0\end{pmatrix}\ \frac{1}{\sqrt6}\begin{pmatrix}1\\1\\2\end{pmatrix}\right)$ and $\Sigma=\begin{pmatrix}\sqrt6&0&0\\0&0&0\end{pmatrix}$.

And, $u_1=\frac{1}{\sqrt6}\cdot\frac{1}{\sqrt3}\begin{pmatrix}3\\3\end{pmatrix}=\frac{1}{\sqrt2}\begin{pmatrix}1\\1\end{pmatrix}$, $u_2=\frac{1}{\sqrt2}\cdot\begin{pmatrix}1\\-1\end{pmatrix}$, ($\sigma_2=\cdots=0$, so u_2 is chosen by $\langle u_1,u_2\rangle=0$)

Now, $A=U\Sigma V^*$, where $U=\begin{pmatrix}u_1&u_2\end{pmatrix}$.

「Eclipse.

Let $T: \mathbb{R}^2 \to \mathbb{R}^2$ be an invertible linear operator, and $S = \{(x,y) \in \mathbb{R}^2 \mid x^2+y^2=1\}$.

Since T is invertible, T has rank 2, therefore it has two singular value:

$\sigma_1 \geq \sigma_2 > 0$ and orthonormal basis $\beta = \{v_1, v_2\}$ and $\beta' = \{u_1, u_2\}$ s.t

$$T(v_1) = \sigma_1 u_1 \quad \text{and} \quad T(v_2) = \sigma_2 u_2.$$

「Polar decomposition.

For any square $A \in M_{n\times n}(\mathbb{C})$, there exists unitary W and semi-positive P such that

$$A = WP.$$

In particular, if A is invertible, this decomposition is unique.

Proof. $A = U\Sigma V^* = \underbrace{UV^*}_{W} \underbrace{V\Sigma V^*}_{P}.$